

Rodrigo David Benitez

Software Engineer · AI/ML Specialist (GenAI, RAG & local/edge inference)

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Summary

Software Engineer and AI/ML Specialist with experience since 2017 spanning applied machine learning, speech/translation technology, and enterprise DevOps. Self-taught; entered the industry as a Spanish–English interpreter and went on to build production speech, translation, and voice-automation software for language-services companies. Core stack: Python/PyTorch, RAG and agentic architectures, model fine-tuning, fully local/edge inference (llama.cpp, GGUF, Ollama), and Jenkins-driven CI/CD with microservices and REST APIs. Led the GenAI team at DigPatho building vision models for breast-cancer diagnosis support; currently DevOps Engineer & AI Implementation at Synopsys supporting developers worldwide.

Skills

- **Programming languages:** Python (expert — ML/AI, Django, FastAPI, Qt/PyQt5/PySide/CustomTkinter, automation) · JavaScript/TypeScript (advanced — Node.js, React) · SQL (advanced — PostgreSQL, MySQL, SQLite incl. FTS5, MongoDB) · Go (intermediate — microservices, APIs) · HTML/CSS (expert)
- **Backend & architecture:** Django, FastAPI, Node.js/Express.js, RESTful APIs, microservices, WebRTC/Socket.io real-time stacks
- **Front end & desktop:** React, Vue.js, HTML5, Bootstrap, CustomTkinter, PyQt5/Qt, Neutralino, WYSIWYG/Markdown editors
- **Databases & storage:** PostgreSQL, MySQL, MongoDB, SQLite (incl. FTS5 full-text), pandas/openpyxl data pipelines
- **Cloud & DevOps:** AWS (incl. AWS Transcribe), Docker, Kubernetes (personal projects), Jenkins CI/CD, Git, PyInstaller frozen builds, machine-bound licensing systems
- **AI/ML — modeling & frameworks:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, GPT-2 fine-tuning, RNNs, GANs, vision transformers, synthetic data generation

- **AI/ML — speech & local inference:** Whisper (STT), MarianMT/OPUS-MT, Coqui TTS, MaryTTS, Vosk, Piper, PaddleOCR, LocateAnything grounding, GGUF/llama.cpp (`llama-server`, Vulkan/CUDA offload), Ollama, custom RAG and agentic memory architectures, embeddings, FAISS vector indexes
 - **Methodologies & tools:** Agile, Scrum, JIRA, Confluence, change management (enterprise SLAs), technical documentation & runbooks
 - **Industry knowledge:** language interpretation & translation operations (bilingual EN/ES) · medical-imaging AI workflows (pathology reporting, deferred diagnosis) · call-center voice automation · corporate developer-support environments (semiconductor/EDA)
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Professional Experience

DevOps Engineer & Automation / AI Implementation — Synopsys Inc.

Buenos Aires, Argentina · Remote · Full-time · Feb 2025 – Present

Supporting a global developer organization at a leading semiconductor and EDA software company.

- Act as DevOps engineer for developer environments: run and troubleshoot **Jenkins CI/CD pipelines** end to end — pipeline triage, job-configuration fixes, build-failure analysis, and handoff to owning teams.
- Design and ship automation that removes manual toil from support and operations: runbook-driven fixes, scripted environment remediation, and self-service automation for recurring developer requests (environment access, permissions, build/toolchain setup).
- Implement **AI-assisted tooling** inside the corporate environment — local-first LLM/RAG helpers for ticket triage, knowledge-base retrieval, and summarization of incidents and runbooks, with no proprietary data leaving the enterprise.
- **Lead the day-to-day team effort:** workload distribution, technical guidance and review of teammates' fixes, and escalation handling.
- Manage project workflows and documentation in **JIRA and Confluence**; authored and standardized runbooks so repeat fixes stop being re-invented by hand.
- Operate inside strict enterprise change-management and security constraints — SLAs, incident documentation, and cross-team escalation.

Tech: Jenkins, JIRA, Confluence, Python automation, local LLM/RAG tooling, enterprise Linux/Windows administration, CI/CD, enterprise change management.

GenAI Team Lead — DigPatho

Buenos Aires, Argentina · Remote · Contract · Jul 2024 – Dec 2024

Led the generative-AI team of a medical/pathology AI venture building vision models for breast-cancer prediction and diagnosis.

- **Led the generative-AI team** developing vision models for breast-cancer prediction and diagnosis — owned the roadmap and model-evaluation strategy; mentored engineers, ran weekly experiment reviews, and coordinated with the clinical team.
- **Deferred-diagnosis tool on a custom RAG framework** — retrieval layer grounds model answers in curated reference material and prior-case context, deferring low-confidence cases instead of blocking the workflow.
- **Semantic description system** — large-context-window transformers that turn image findings into standardized, clinician-readable patient reports, keeping phrasing consistent with pathology reporting conventions.
- **Integrated vision models into diagnostic workflows** with explicit deferral paths for low-confidence cases.
- Created end-to-end diagnostic workflows for deferred medical diagnosis — from image ingestion to finding extraction to the human review step.

Tech: Python, PyTorch/TensorFlow (vision transformers), RAG, large-context transformers, research-model optimization, medical-imaging pipelines.

Freelance AI/ML Developer — Self-Employed

Remote · International clients · Self-employed · Jun 2022 – Jul 2024

Independent deep-learning and applied-AI development for international clients across industries, focused on automation that reduces operational cost — shipping local-first models that run without paid external APIs.

Deliverables:

- **Multilingual transcription platform** — scaled to 1,000+ concurrent audio streams with accuracy of up to 95% across 20 languages; AWS Transcribe + custom per-language ML models (stream-oriented ingestion, per-language routing, fallback chain).
- **VoIP AI interpreter** — AI-based interpretation and live transcription layered onto VoIP calls.
- **AI agentic voice assistant** — corporate staff assistant with voice interaction and agent-style task execution over internal systems (tool-calling loop with an STT/TTS front end).

- **Remote Screen Caster** — high-performance screen sharing over WebRTC + Socket.io with end-to-end encryption for low-latency remote support.
- Plus voice-cloning translation, real-time multilingual chat, and call-center voice-autoresponder systems for additional clients.

Impact: operational-cost reductions of **up to 30%** through AI-powered automation across multiple engagements; on one platform, cloud-computing spend fell by **up to 45%** via right-sizing, local-inference offload, and caching.

Technical scope across engagements:

- Custom ML model development in **PyTorch** (TensorFlow/scikit-learn where client stacks required it): RNNs for sequence problems, GANs for synthetic data, **GPT-2 fine-tuning** for conversational and text-generation tasks.
- Microservices architecture with cost-conscious deployment — right-sizing, local-inference offload, and caching.

IT Specialist & Developer — Homeland Language Services

Oxnard, California, USA · Remote · Full-time · May 2021 – Jun 2022

Hired as a Spanish–English interpreter and promoted into corporate IT; grew into the company’s internal developer across websites, automation tooling, IT support, and voice-AI experiments.

- Promoted from interpreter to corporate IT within **two months** of hiring.
- Provided IT support and small-to-medium programming for a remote-first organization — end-user support, hardware/software troubleshooting, and onboarding of remote staff across time zones.
- Developed internal websites, reporting scripts, and workflow automations in Python that removed manual steps from daily operations.
- Created custom solutions improving operational efficiency across departments (scheduling helpers, status-report generators, internal portals).
- Built and maintained a **virtual interpreter chatbot with voice** (GPT-2, Whisper, MaryTTS, Qt).
- Acted as the internal bridge between the interpretation business and engineering: understood the interpreter’s problem and built the fix.

Tech: Python, JavaScript, Django/FastAPI, GPT-2, Whisper, MaryTTS, Qt, general IT tooling.

Software Engineer — Elite Language Services

Fort Lauderdale, Florida, USA · Remote · Full-time · Jan 2019 – Feb 2021

Joined as a Spanish–English interpreter and transitioned into software development within the same engagement, automating the workflows used on the floor. Every tool below addresses a bottleneck observed directly during interpretation work.

- **Live translation tools for interpreters** — real-time speech-to-text plus translation assist on interpreter workstations (Whisper + GPT-2 assisted rendering, Qt interface), Python.
- **Macro shortcut tool** — Selenium + OpenCV + PyAutoGUI automation executing multi-step workflows from a single keystroke; Python and JavaScript.
- **Keyword finder add-on** — companion to the live-translation tool that scans ongoing discourse and surfaces definitions/usage of selected words on demand, giving interpreters instant terminology support.
- **Caption tools and time organizers** — converts interpreted audio into timed captions and keeps session calendars organized.
- **AutoHotkey automation suite + auto-clickers** — lightweight optimization of repetitive interpreter desktop tasks (window switching, clipboard staging, call-control shortcuts).
- **Web-browsing bots for internal process automation** — Selenium/OpenCV/PyAutoGUI bots that replaced manual browser drudgery in back-office operations.
- **Global internal database with unique-hash multi-level authentication** — Django + FastAPI backend over a shared company database, pandas/numpy processing, Bootstrap front end; each access tier verified by a unique hash.
- **On-demand interpretation VoIP platform** — Bootstrap + Django + Twilio + React front end with SQL persistence; interpreters and clients connect over VoIP with structured session handling.
- **CSV / XLSX bidirectional format converter** — pandas/openpyxl utility for instant, lossless spreadsheet interchange between client formats.

Impact: the multilingual translation tooling of this period increased interpreter efficiency by up to 40%.

Tech: Python, JavaScript, Whisper, GPT-2, Qt, Selenium, OpenCV, PyAutoGUI, AutoHotkey, Django, FastAPI, Twilio, React, Bootstrap, pandas, openpyxl, SQL.

Junior Developer / Project Manager — Dialog One

Saint Paul, Minnesota, USA · Remote · Contract · Aug 2018 – Jan 2019

Developer and project manager for a US language-services company, owning both the technical build and the delivery side of a voice-enabled virtual interpreter: a desktop application letting an interpreter (or a monolingual end user) interact with a GPT-2-driven conversational agent through speech.

- Wore two hats: implemented the Python voice pipeline and ran the project (requirements, milestones, stakeholder communication) as PM.
- Built a speech-driven conversational loop — audio capture → Whisper speech-to-text → GPT-2 text generation → MaryTTS speech synthesis — orchestrated inside a Qt desktop client for Spanish/English interpretation-oriented interaction.
- Defined the product spec with interpretation-domain stakeholders and managed expectations around what a 2018-era GPT-2 model could and could not do.
- Negotiated scope, contracts, and payment milestones for a client-funded build under fixed-bid constraints.

Tech: Python, GPT-2, OpenAI Whisper (early adoption), MaryTTS, Qt.

Junior Developer (Intern) — Byron RPS

Alcudia de Carlet, Valencia, Spain · Remote · Internship · Jan 2017 – Jun 2018

First professional entry into software development as a self-taught programmer: a real codebase, real team rituals, and real production constraints.

- Implemented data-processing and search algorithms from specification for internal tools — parsing, sorting, and normalization routines over structured datasets.
 - Worked inside a shared Python codebase with weekly code reviews, clean-code habits, documentation, git workflows, and release discipline.
 - Early hands-on with **AI integration** — recurrent and convolutional architectures on prototype-stage experimental features.
 - Participated in splitting a monolith into small, independently deployable **microservices** and the HTTP/API contracts between them.
 - Learned project management: task breakdown, estimation, progress tracking, and reporting.
 - Primary language of daily work: Python and its frameworks.
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Selected Projects

Local-first, offline-capable Python tooling. Public repositories on github.com/RodrigoBenitez343.

- **Arrow** — *Generalistic RPA for Edge Devices (private — demo on request)*. Node-graph RPA for offline edge devices — users wire nodes (triggers, screen/UI actions, OCR, vision, branching) instead of coding; runs unattended with fully local inference. No API calls, no paid resources.
 - **[CallScribe v2](#)** — *Real-time bilingual call interpreter*. Live local transcription for bilingual interpreters: Whisper STT + MarianMT translation, color-coded transcript, click-to-translate; 30 languages per call side, machine-bound licensing.
 - **Other local-first AI products** — [OnboardAI](#): 100% offline screen-aware assistant (OCR → local GGUF LLM, grounded chat); [CoVe](#): research & synthesis workbench (web/arXiv + FAISS + iterative LLM chains); [Documenter](#): self-generating code docs via local Ollama models; [CibernetDocs](#): Markdown → A4 PDF formatting pipeline (web editor + desktop app); [LocateAnything video pipeline](#): text-query labeling of videos on local llama.cpp.
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Education

- **Software Development** — 100% self-taught developer (no formal CS degree); learning path built on shipped products rather than coursework.
 - **Language-Interpretation Training** — professional Spanish–English interpretation practice (worked as interpreter before and while becoming a developer).
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Languages

- Spanish — Native
- English — Full professional proficiency